1. $h(t) = e^{-2t}\, u(t) \qquad f(t) = e^{-t}\, u(t)$

$$y(t) = \int_0^t e^{-\tau}\, e^{-2(t-\tau)}\, d\tau \quad t \geq 0 = \left(e^{-t} - e^{-2t}\right) u(t)$$

2. $h(t) = 6e^{-t}\, u(t) \qquad$ (a) $2\, u(t) \quad$ (b) $3e^{-3t}\, u(t)$

and (a) $12(1 - e^{-t})\, u(t) \quad$ (b) $9(e^{-t} - e^{-3t})\, u(t)$

(3) $h(t) = \left(2e^{-2t} - e^{-t}\right) u(t) \qquad f(t) = 10e^{-3t}\, u(t)$

$$y(t) = \frac{20}{-3-(-2)}\left(e^{-3t} - e^{-2t}\right) u(t) - \frac{10}{-3-(-1)}\left(e^{-3t} - e^{-t}\right) u(t)$$

$$= \left(-5e^{-t} + 20e^{-2t} - 15e^{-3t}\right) u(t)$$

4. $h(t) = e^{-2t}\, u(t) \qquad f(t) = \sin 3t\, u(t)$

$$y(t) = \frac{1}{13}\left[3e^{-2t} + \sqrt{13}\cos\left(3t - 146.32^\circ\right)\right] u(t)$$

$$= \frac{1}{13}\left[3e^{-2t} - \sqrt{13}\cos\left(3t + 33.68^\circ\right)\right] u(t)$$

5. Consider discrete LTI system with unit ~~sample~~ response $h[n] = (n+1)\alpha^n u[n]$ where $|\alpha| < 1$, show that the step response of this system is

$$S[n] = \left[\frac{1}{[\alpha-1]^2} - \frac{\alpha}{(\alpha-1)^2}\alpha^n + \frac{\alpha}{[\alpha-1]}(n+1)\alpha^n\right] u[n].$$

hint $\sum_{k=0}^{N} (k+1)\alpha^k = \frac{d}{d\alpha}\sum_{k=0}^{N+1} \alpha^k$

Ans. $S[n] = \sum_{k=-\infty}^{n} h[k] = \sum_{k=-\infty}^{n} (k+1)\alpha^k u[k] = \sum_{k=0}^{n} (k+1)\alpha^k u[k] = \frac{d}{d\alpha}\left[\frac{1-\alpha^{n+2}}{1-\alpha}\right]$

$$S[n] = \left[\frac{1-(n+2)\alpha^{n+1}}{1-\alpha} + \frac{1-\alpha^{n+2}}{(1-\alpha^2)}\right]; \; n > 0 = \left[\frac{1-(n+1)\alpha^{n+1} - \alpha^{n+1}}{1-\alpha} - \frac{1-\alpha^{n+2}}{1-\alpha^2}\right] u[n]$$

$$= \left[\frac{1}{(\alpha-1)^2} - \frac{\alpha^{n+1}}{(\alpha-1)^2} + \frac{\alpha^{n+1}(n+1)}{(\alpha-1)}\right] u[n]$$

6. a) $x(t) \rightarrow \boxed{h(t)} \rightarrow \boxed{g(t)} \rightarrow y(t)$ $\quad x(t) = e^{-t}$, $h(t) = e^{-t}u(t)$, $g(t) = u(t) + \delta(t)$

find $y(t)$

b) $x[n] = \left[\frac{1}{2}\right]^n$, $h[n] = \left[\frac{1}{2}\right]^n u[n]$, $g[n] = \delta[n] - \frac{1}{2}\delta[n-1]$

find $y[n]$

$x[n] \rightarrow \boxed{h[n]} \rightarrow \boxed{g[n]} \rightarrow y[n]$

7. Consider the cascade of 2 systems. A is an LTI system. The second system B is known to be the inverse of system A

$x(t) \rightarrow \boxed{A} \xrightarrow{y(t)} \boxed{B} \rightarrow x(t)$

a) What is the response of system B to the i/p $a y_1(t) + b y_2(t)$ where a & b are const.

b) What is the response of system B to the i/p $y_1(t-2)$?

8. $x(t) = u(t+0.5) - u(t-0.5)$ $\quad h(t) = e^{j\omega_0 t}$

a) Det a value of ω_0 which ensures that $y(0) = 0$

b) Is your answer to a part unique?

$\omega_0 = 2\pi$, No

9. Check the system causality & stability

$h(t) = e^{-6t} u(3-t)$ NU

$h(t) = e^{2t} u(-1-t)$ NS

$h(t) = \left[2e^{-t} - e^{\frac{(t-100)}{100}}\right] u(t)$ CU

$h(t) = t e^{-t} u(t)$ CS

$h[n] = \left(\frac{1}{3}\right)^n n\, u[n-1]$ CS

$h[n] = (5)^n u[3-n]$ NS

$h[n] = \left(\frac{1}{2}\right)^n u[-n]$ AU

2.22 a)

$x(t) = e^{-\alpha t}\, u(t) \qquad h(t) = e^{-\beta t}\, u(t)$

Replace $t \to (t-\tau)$

$$y(t) = \int_{-\infty}^{\infty} e^{-\alpha\tau} u(\tau)\, e^{-\beta(t-\tau)} u(t-\tau)\, d\tau$$

$$= \int_0^t e^{(\beta-\alpha)\tau} e^{-\beta t}\, d\tau$$

$\alpha = \beta \qquad t e^{-\beta t} u(t)$

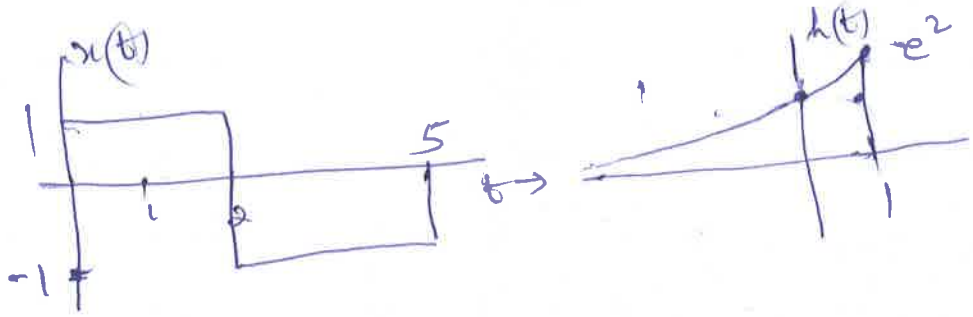
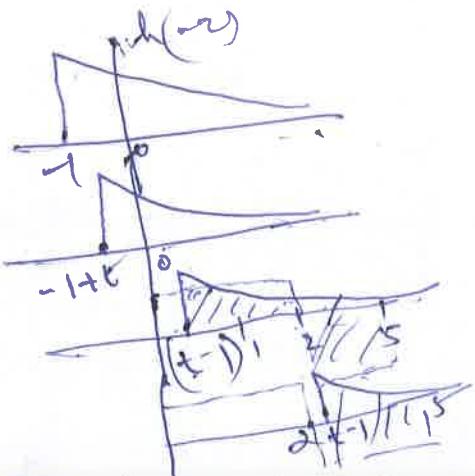
$\alpha \neq \beta \qquad \dfrac{e^{-\beta t}\left[e^{(\beta-\alpha)t} - 1\right]}{\beta - \alpha} u(t)$

b)

$x(t) = u(t) - 2u(t-2) + u(t-5) \qquad h(t) = e^{2t} u(1-t)$

$$y(t) = \int_{-\infty}^{\infty} x(\tau)\, h(t-\tau)\, d\tau = \int_0^2 h(t-\tau) - \int_2^5 h(t-\tau)\, d\tau$$

$$\int_0^2 e^{2(t-\tau)}\, d\tau - \int_2^5 e^{2(t-\tau)}\, d\tau = \qquad t \leq 1$$

$$\int_{t-1}^2 e^{2(t-\tau)}\, d\tau - \int_2^5 e^{2(t-\tau)}\, d\tau = \qquad 1 \leq t \leq 3$$

$$-\int_{t-1}^5 e^{2(t-\tau)}\, d\tau = \frac{e^{2t-10} - e^2}{2} \qquad 3 \leq t \leq 6$$